

Prathmesh Waghmare

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SUMMARY

B.Tech (Artificial Intelligence & Machine Learning) student with hands-on Research & Development experience at the Space Applications Centre (SAC), ISRO, building deep-learning and computer-vision models — including YOLO, R-CNN, and RNN-based architectures — for real-time object detection and denoising of satellite imagery. Proficient in Python, PyTorch, TensorFlow, and OpenCV, with applied project experience spanning satellite image analysis, healthcare AI, and full-stack web development. Author of three IEEE-indexed research publications. Proven leadership track record as Vice President (IGNITRA 3.0), Event Management Lead (GDG on Campus DMIHER), and General Secretary (Student Council).

EDUCATION

Datta Meghe Institute of Higher Education and Research (DMIHER) — Expected 2027

Bachelor of Technology (B.Tech) — Artificial Intelligence & Machine Learning, Wardha, Maharashtra

CGPA: 7.52 / 10 (through 6th Semester)

TECHNICAL SKILLS

Programming: Python, JavaScript

Machine Learning: Supervised Learning, Model Training, Evaluation & Testing, R-CNN, RNN

Deep Learning & CV: PyTorch, TensorFlow, OpenCV, CNN, YOLO (v8, v11, v26), Faster R-CNN, RNN, Mamba, BDH Architectures, Image Denoising & Enhancement, Image Preprocessing & Annotation

Tools: Google Colab, Roboflow, LabelImg, Git, Claude, GPT

Web Technologies: HTML, CSS, React.js, Node.js

EXPERIENCE

Research Intern — Space Applications Centre (SAC), ISRO

6-Month On-Site Research Internship, Ahmedabad, Gujarat | Apr 2026 – Oct 2026 (Ongoing)

- Engineered real-time object-detection pipelines for satellite imagery using multiple YOLO variants (YOLOv8, YOLOv11, YOLOv26), Faster R-CNN, and RNN-based sequence models in Python, PyTorch, and OpenCV.
- Researched and prototyped Dragon Hatchling (BDH) and Mamba-based architectures to improve the efficiency of large-scale satellite image analysis.
- Built image-denoising and enhancement modules to remove noise from real-time satellite data, improving downstream model accuracy.
- Performed image preprocessing and annotation (Roboflow, LabelImg) on remote-sensing datasets to support high-quality model training.
- Collaborated with cross-functional research teams on geospatial and AI/ML initiatives under SAC-ISRO.

Web Development Intern — Cosmos Digital

On-Site Internship, India | Apr 2025 – Jun 2025

- Built a business-oriented web application and stakeholder dashboard using HTML, CSS, JavaScript, and Node.js.
- Implemented responsive, user-friendly features and partnered with the team to translate business requirements into functionality.

PROJECTS

Real-Time Object Detection from ISRO Satellite Imagery — *Python, PyTorch, OpenCV, YOLOv8/v11/v26*

Apr 2026 – Present

Trained and benchmarked multiple YOLO model variants for real-time detection of objects in satellite images from ISRO datasets; optimized the inference pipeline for real-time performance.

Real-Time Satellite Image Denoising & Enhancement — *Python, OpenCV, Deep Learning*

Apr 2026 – Present

Designed denoising models to remove noise from real-time satellite imagery, improving image clarity for downstream object-detection tasks.

COVID-19 & Pneumonia Detection from Chest X-Ray Images — *PyTorch, CNN*

Feb 2025 – Mar 2025

Built a 4-class deep-learning classifier (COVID-19, Pneumonia, Viral Infection, Normal) for real-time chest X-ray screening.

Real-Time Labour Booking Platform for Farmers — *React.js, Node.js, JavaScript*

Dec 2025 – Jan 2026

Built a real-time web platform enabling farmers to book agricultural laborers on demand, with live availability and instant booking confirmation.

RESEARCH PUBLICATIONS

- “A Review on the Diabetes Treatment using Artificial Intelligence and Machine Learning” — 8th ICICT 2025, Tribhuvan University, Nepal. Published, IEEE Xplore.
- “Assessing ROM with AI Photography: Validity & Application” — 6th ICESC 2025, Coimbatore, India. Published, IEEE Xplore.
- “Smart Water Irrigation Using IoT, Artificial Intelligence, and Machine Learning Techniques” — 7th ICMCSI 2026, Purbanchal University, Nepal. Accepted for Presentation, IEEE.
- “Real-Time Web-Based Marketplace for On-Demand Matching of Agricultural Labor” — ICCSECD 2026. Accepted for Presentation, IEEE.

LEADERSHIP & CERTIFICATIONS

Leadership: Vice President — IGNITRA 3.0; Event Management Lead — GDG on Campus DMIHER; General Secretary — Student Council

Certifications: Data Science & Analytics (HP LIFE, 2025); AI/ML for Geodata Analysis (IIRS Dehradun, 2024); Data Analysis with Python (Coursera, 2024); Introduction to Machine Learning on AWS (2023); Soft Skill Development (NPTEL, IIT Kharagpur, 2025)

Hackathon: Bharatiya Antariksh Hackathon — ISRO (2024), Participant